

1. Find the average price of pastries for each category from the pastries table.

```
select category, count(name),  
       avg(price)  
from pastries  
group by category;
```

Result Grid	Filter Rows:	Export:
category	count(name)	avg(price)
Pastry	3	4.666666666666667
Muffin	2	3.625
Bakery	1	4.25
Scone	2	3.85
Breakfast	2	8
Cake	2	3.875

2. Find the total number of baristas at each experience level from the baristas table.

```
select experience_level, count(experience_level)  
from baristas  
group by experience_level;
```

Result Grid	Filter Rows:
experience_level	count(experience_level)
Junior	2
Senior	2
Mid-Level	2

3. Count the total number of shops located in each city from the shops table.

```
select city, count(city)  
from shops  
group by city;
```

Result Grid			Filter Rows:
	city	count(city)	
▶	Seattle	1	
	Portland	1	
	Austin	1	
	Chicago	1	
	Denver	1	
	Boston	1	

4. Find the maximum price among pastries for each category from the pastries table.

```
select category, MAX(price)
from pastries
group by category;
```

Result Grid			Filter Rows:
	category	MAX(price)	
▶	Pastry	5.5	
	Muffin	3.75	
	Bakery	4.25	
	Scone	3.85	
	Breakfast	8.5	
	Cake	4	

5. Count how many pastries have been added by each shop using the shopID column from the offers table

```
select shopID, count(pastryID)
from offers
group by shopID
```

Result Grid			Filter Rows:
	shopID	count(pastryID)	
▶	1	2	
	2	2	
	3	2	
	4	2	
	5	2	
	6	2	

6. Find the name, category, and price of any pastry whose price matches the maximum price within its category.

```
select name, category, price
from pastries x
where price = (
    select MAX(price)
    from pastries
    where category = x.category
);
```

	name	category	price
▶	Almond Croissant	Pastry	5.5
	Blueberry Muffin	Muffin	3.75
	Cinnamon Roll	Bakery	4.25
	Chocolate Scone	Scone	3.85
	Blueberry Scone	Scone	3.85
	Breakfast Burrito	Breakfast	8.5
	Lemon Loaf	Cake	4

7. Find the unique shop IDs from the offers table that have offered at least one pastry whose price is strictly greater than the overall average price of all pastries.

```
select distinct x.shopID
from offers x
join pastries y
    on x.pastryID = y.pastryID
where y.price > (
    select AVG(price)
    from pastries
);
```

Result Grid	
	shopID
▶	1
	5

8. Find the shop ID and pastry ID for the records in the offers table that have the earliest **date_added** (minimum date).

```
select shopID, pastryID
from offers
```

```

where date_added = (
  select MIN(date_added)
  from offers
);

```

Result Grid			Filter Rows:
	shopID	pastryID	
▶	1	1	

9. Find the shop ID(s) that offer the highest number of pastries, utilizing a subquery to evaluate the maximum count per shop.

```

select shopID
from offers
group by shopID
having count(pastryID) = (
  select MAX(pastry_count)
  from (
    select count(pastryID) as pastry_count
    from offers
    group by shopID
  ) as shop_counts
);

```

Result Grid		
	shopID	
▶	1	
	2	
	3	
	4	
	5	
	6	

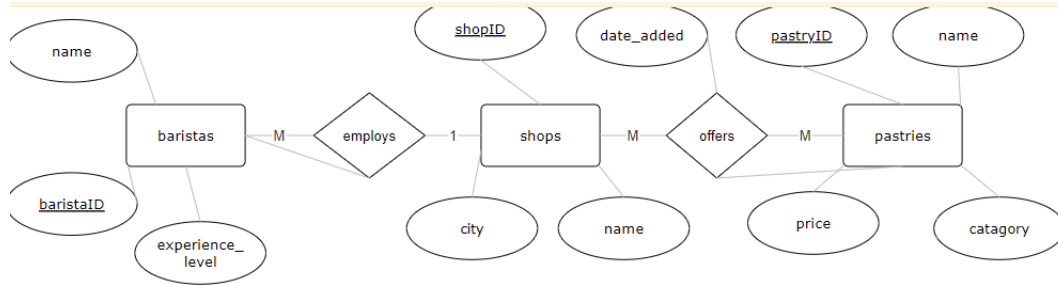
10. Find the names of baristas who work at shops located in 'Seattle' using nested subqueries.

```

select name
from baristas
where baristaID in (
  select baristaID
  from employs
  where shopID in (
    select shopID
    from shops
    where city = 'Seattle'
  )
);

```

Result Grid	Filter
name	
▶ Emma Watson	
Liam Chen	



Justification for full participation of baristas and pastries for shops:

- A barista can only be employed if they are working at a shop, therefore all baristas must be in an employs relationship with shops.
- A pastry can only exist in a shop's system if they can offer it, therefore all pastries must participate in the offers relationship.